

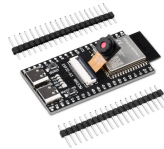
DIY - CYBERSHOT

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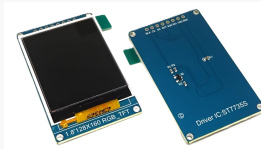
Assembly guide for the CyberShot Cam — DIY digital camera built with ESP32-S3, live viewfinder, long exposure, databending effects and web gallery.

WHAT YOU'LL NEED

- 01** **Esp32-s3 Cam Ov5640 Wifi Bluetooth 5.0
Ble N16r8**



- 02** **TFT LCD Display 1.8" 128x160 ST7735**



- 03** **Analog Joystick Module KY-023**



- 04** **Push Button (momentary)**



- 05** **Micro SD Card / Any model**



- 06** **5mm High-Brightness LED, Red (optional)**



You'll also need: soldering iron / solder / wires for connections or a circuit board.

PINOOTS

TFT Display (ST7735)

Connected via hardware SPI (FSPI bus). Keep wires short for reliable high-speed operation.

TFT Pin	Function	ESP32-S3 GPIO
VCC	Power	3.3 V
GND	Ground	GND
SCK	SPI Clock	GPIO 47
SDA (MOSI)	SPI Data	GPIO 45
CS	Chip Select	GPIO 43
DC	Data/Command	GPIO 14
RST	Reset	GPIO 21

NOTE: the TFT uses the FSPI bus. Do not share this bus with other SPI devices.

Analog Joystick (KY-023)

Joystick Pin	ESP32-S3 GPIO
VRx (X axis)	GPIO 1 (ADC)
VRy (Y axis)	GPIO 2 (ADC)
SW (button)	GPIO 42
VCC	3.3 V
GND	GND

Push Button (momentary)

Button Pin	ESP32-S3 GPIO
Terminal A	GPIO 41
Terminal B	GND

NOTE: internal pull-up enabled, active LOW — no external resistor needed.

■ LED (system indicator)

LED Terminal	Connection
Anode	100 Ohm resistor -> GPIO 48
Cathode	GND

SOFTWARE SETUP

- | | | |
|----|-----------------------------|--|
| 01 | Install PlatformIO | VS Code extension or CLI — https://platformio.org |
| 02 | Clone the repository | <code>git clone https://github.com/seu-usuario/cybershot-cam</code> |
| 03 | Open in VS Code | Open the project folder with the PlatformIO extension active |
| 04 | Connect the board | Connect the ESP32-S3 via USB-C; hold BOOT on the first flash |
| 05 | Flash the firmware | Click "Upload" in PlatformIO, or run: <code>pio run --target upload</code> |
| 06 | Prepare the SD card | Insert a FAT32-formatted microSD card before the first boot |
| 07 | Configure WiFi | On the device: MENU -> WIFI -> CONFIGURE |

Dependencies

Automatically installed by PlatformIO via `platformio.ini` — no manual setup needed.

- + adafruit/Adafruit ST7735 and ST7789 Library
- + adafruit/Adafruit GFX Library
- + bitbank2/JPEGDEC

CONTROLS & HOW TO USE

Input controls

Action	Result
HOLD button (> 800 ms)	Open / close the main menu
PRESS button (short)	Take a photo from the viewfinder
JOYSTICK UP / DOWN	Navigate menu items
JOYSTICK LEFT / RIGHT	Confirm / adjust values
JOYSTICK PRESS (SW)	Back / cancel

Main menu (hold the button in viewfinder)

Item	Function
VF COLOR	Viewfinder color palette (green / red / pink / white / cyan)
LONG EXP	Long exposure with frame stacking — OFF / 3S / 5S / 10S
TIMER	Self-timer before capture — OFF / 3S / 5S / 10S
EFFECTS	JPEG databending: ZIGZAG PERM, DQT EROSION, DHT REMAP, SCAN SWAP, CHROMA AMP
WIFI	Network configuration submenu (see below)
FORMAT SD CARD	Format the SD card — confirm with joystick
EXIT	Close the menu and return to viewfinder

Viewfinder indicators

Indicator	Meaning
L3S / L5S / L10S	Long exposure active (3s / 5s / 10s stacking)
EV bar	Exposure compensation (-3 to +3)
Color palette	Viewfinder visual style — does not affect saved photo

■ WiFi & web gallery

1. MENU -> WIFI -> CONFIGURE: opens the **CyberShot-Setup** access point
2. Connect your phone to that network and open a browser at **192.168.4.1**
3. Enter your home WiFi credentials and save
4. Browse the gallery at **http://cybershot.local** (or the IP shown on screen)

■ WIFI submenu

Option	Function
CONFIGURE	Opens a captive portal to set up a new WiFi network
CONNECT STA	Connect using already saved credentials
DIRECT AP	Create open access point "CYBERSHOT" — no password
DISCONNECT	Disconnect from current WiFi
BACK	Return to main menu

TIPS & NOTES

- **PSRAM required** - The project uses PSRAM extensively. Make sure your ESP32-S3 board has at least 8 MB of PSRAM — boards without PSRAM will crash when allocating frame buffers.
- **Long exposure** - Stacks multiple JPEG frames (3s / 5s / 10s) for a well-exposed result. The LED stays on throughout the capture. Works best with the camera on a stable surface.
- **Photo resolution** - XGA 1024 x 768 JPEG for both normal and long exposure shots. Saved as /PHOTO_XXXX.JPG at the root of the SD card.
- **Web gallery** - At <http://cybershot.local>: browse, edit (brightness, contrast, saturation, vignette, chromatic aberration, copy fx, pixel sort) and download photos. Runs entirely on the ESP32 — no internet required.
- **WiFi credentials** - Stored in ESP32 NVS flash memory. Never written to source code or the SD card.
- **Format SD card** - MENU -> FORMAT SD CARD. Confirm with the joystick. FAT32 only, max. 32 GB.
- **Power** - Runs via USB-C (5 V). For portable use: 3.7 V LiPo battery with a charge/boost module (TP4056 + MT3608 or similar).
- **White screen** - If the display stays white, check the SCK / SDA / CS / DC / RST wiring. RST must be pulsed LOW then HIGH on startup.

DIY - CYBERSHOT // open source project // built with PlatformIO + esp32-camera // github.com/seu-usuario/cybershot-cam